

What type of tie is being described? Is it symmetric, asymmetric, or anti-symmetric? If it's asymmetric, is it described as being reciprocal? If it is not clearly reciprocal, assume it is not. Is the connection type based on a flow, interaction, social relation, or similarity?

Symmetry Reciprocal? Type

- 1) Omar is Isaac's boss. **Anti-symmetric | non-Reciprocal | Social Relation**
Isaac cannot be Omar's boss by definition, thus it is anti-symmetric and therefore non-reciprocal. Boss is a social relation.
- 2) Said ignores Tom's FB friend request. **Symmetric | Reciprocal | Social Relation**
The FB request is the tie type, and FB is inherently symmetric and thus reciprocal. Friend is generally a social relation.
- 3) Wikipedia references another website. **Asymmetric | non-Reciprocal | Flow**
The other website could reference Wikipedia back, it's an open question, so asymmetric. Although it's described as non-reciprocal and the reference can be thought of as the flow of information, but it's not the easiest type.
- 4) Jeff and Neil email each other at work. **Asymmetric | Reciprocal | Interaction/Flow**
Also tricky. So, if emailing is optional, then it's asymmetric. While if the relation is "email each other" then it's a symmetric tie. In both cases since it's "each other" it's reciprocal, and it's either an interaction or the flow of emails.
- 5) Denzel is Greg's advisor. **Anti-Symmetric | non-Reciprocal | Social Relation**
Depending on if you think the flow of advice could go both ways or not, could be asymmetric. Generally, the title "advisor" though means they give you advice and you don't give advice to them, so with this it'd be non-reciprocal.
- 6) Lu and Beverly hang out sometimes. **Symmetric | Reciprocal | Interaction**
Hanging out is generally something you have to do together, so it's symmetric and therefore reciprocal. It's an interaction as it's an activity.
- 7) Smriti and Kevin don't know each other. **Symmetric | Reciprocal | Social Relation**
Depends on how you want to think of "knowing each other." It's odd if I know you but you don't know me. This is more of "knowing" than "knowing of", so it'd be symmetric, and therefore reciprocal. The type is a bit unclear so let's say social relation.
- 8) Michael thinks of Willie as a friend. **Asymmetric | non-reciprocal | Social relation**
There is no mention that Willie needs to also think of Michael as a friend, so it's asymmetric. The reciprocity is an open question. It's also non-reciprocal because it doesn't say he does think of Michael as a friend. Friend is a social relation.
- 9) Ricardo and Youssef both like pizza. **Symmetric | Reciprocal | Similarity**
The relation is liking pizza. This is inherently symmetric as you can't be in a relation if one person also doesn't like pizza. It's therefore reciprocal, and the tie is based on a similarity of liking pizza.
- 10) Flights occur between Denver and Houston. **Asymmetric | Reciprocal | Flow**
So this is also a bit hard. You can imagine that the relation of "flights" don't have to go both ways. Like flights could only go from Denver to Houston and not back. This would make it asymmetric. But they do go both ways, so it's reciprocal. If you read the relationship as "flights occur between" then you have a point for arguing it's symmetric and thus reciprocal. Planes are a flow.

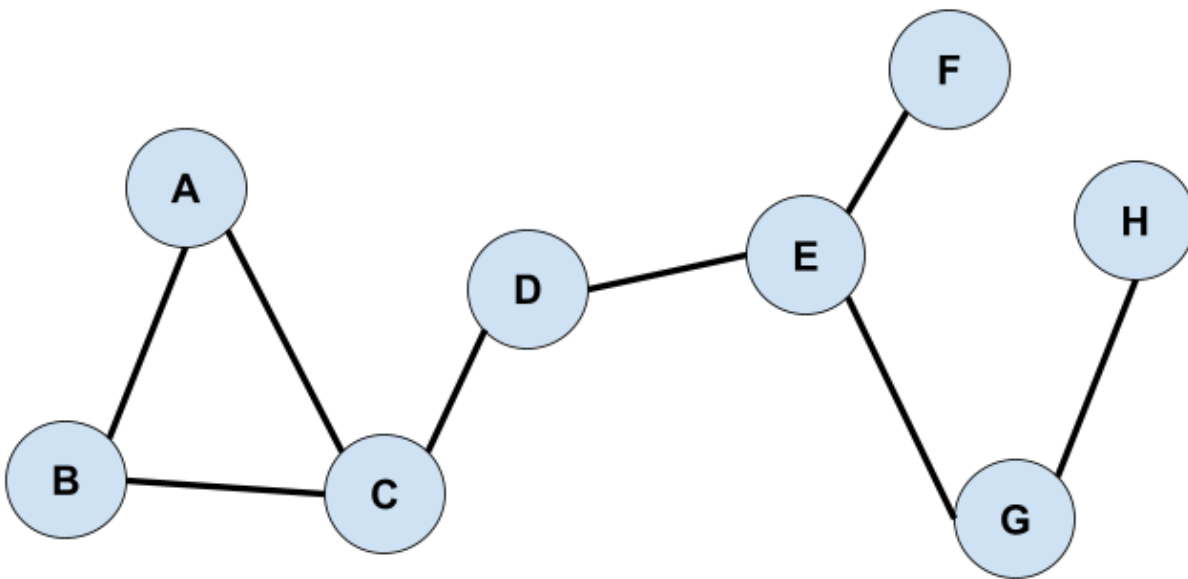
Is the described tie Strong or Weak? Identify at least two elements of time, frequency, intimacy, or exchange are helping to drive your decision.

- 1) You see the same person at the coffee shop every day. In your mind you named them Alex and gave them a whole life story. You've never talked, but you feel like you know them.

Frequency is high, but the intimacy is low and there is no exchange. Weak at best, might not even really be a tie.

- 2) You haven't talked to your best friend from college in a while, but you both have jobs and were busy. They got you through a bad breakup and you lived together senior year. Strong, but there are good reasons to be weak depending on how much time has elapsed. The level of intimacy and exchange were high, and you may have been friends for quite some time, but the longer this tie goes inactivated, the weaker it might become (a lower frequency of friend activation).

Create an adjacency matrix reflecting the below graph.



	A	B	C	D	E	F	G	H
A	--	1	1	0	0	0	0	0
B	1	--	1	0	0	0	0	0
C	1	1	--	1	0	0	0	0
D	0	0	1	--	1	0	0	0
E	0	0	0	1	--	1	1	0
F	0	0	0	0	1	--	0	0
G	0	0	0	0	1	0	--	1
H	0	0	0	0	0	0	1	--

What type of matrix is the above matrix?

Symmetric matrix because the graph is an undirected graph.

What type of relations might be captured by the above matrix?

Are A and B siblings? Are A and B roommates? Co-workers?

What are the logical principles that underlie Balance Theory?

The basic two principles are “entities I like should also like each other” and “entities I don’t like should like each other.” These two logics then can get extended into 4 rules we’d expect.

What is the theoretic concept that seeks to explain why Balance Theory doesn’t always work?

Constraint

What happens to the world when balance theory is perfectly correct?

With perfect constraint, networks split into two clusters where within each cluster everyone likes each other, but does not like actors in the other cluster.